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Machine learning for predicting outcomes, complications and resource utilisation after hip arthroscopy: A systematic review.

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CLINICAL & AGENTIC AI TAKEAWAY

Machine learning models in hip arthroscopy show variable performance in predicting outcomes and complications, with most studies reporting no significant difference to traditional regression.

ABSTRACT

Purpose: Machine learning (ML) algorithms are increasingly used to predict outcomes in orthopaedic surgery, but their utility in hip arthroscopy remains unclear. This study aimed to (1) evaluate ML models for predicting outcomes, complications, and resource use following hip arthroscopy, (2) compare ML performance to traditional statistical methods, and (3) critically appraise the quality of existing literature. Methods: A systematic search of PUBMED, MEDLINE, EMBASE and the Cochrane Central Register of Controlled Trials was conducted on 4 April 2023. Included studies used ML models for clinical prediction in hip arthroscopy. Data on study design, outcomes, model development, validation and adherence to TRIPOD guidelines were extracted. Results: Seventeen studies involving 37,549 patients (52.8% female; mean age 34.4 ± 6.1 years) were included. ML was used to predict clinical outcomes (47.1%), adverse events (47.1%) and resource utilisation (5.9%). The median area under the curve was 0.66 (range 0.5-0.94) for clinical outcomes and 0.7 (range 0.6-0.9) for adverse events. Resource utilisation was reported with a root-mean-square error of $3800, a logarithmic error of 0.2, and R^2 of 0.8. Only four studies compared ML to traditional regression; two found no difference, while two favoured ML. On average, studies met 12.3 of the 22 TRIPOD criteria, with only 29.4% meeting more than two-thirds. Conclusion: ML models in hip arthroscopy show variable performance in predicting outcomes and complications. Furthermore, most studies comparing ML models to traditional regression did not report any significant difference. Lastly, variability in adherence to TRIPOD guidelines highlights the need for future studies to improve transparency and reporting in the development of ML algorithms for hip arthroscopy. Given the potential of ML models in enhancing clinical decision-making for hip arthroscopy, along with the findings of the current study, their application should be used with caution. Level of evidence: Level IV, systematic review of level II-IV studies. © 2026 European Society of Sports Traumatology, Knee Surgery and Arthroscopy.

Keywords: Artificial intelligence, Machine Learning, Hip Arthroscopy, Predictive Modelling, Deep Learning

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: systematic review. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Machine learning models in hip arthroscopy show variable performance in predicting outcomes and complications, with most studies reporting no significant difference to traditional regression.

Extended Repository Abstract

Purpose Machine learning (ML) algorithms are increasingly used to predict outcomes in orthopaedic surgery, but their utility in hip arthroscopy remains unclear. This study aimed to (1) evaluate ML models for predicting outcomes, complications, and resource use following hip arthroscopy, (2) compare ML performance to traditional statistical methods, and (3) critically appraise the quality of existing literature. Methods A systematic search of PUBMED, MEDLINE, EMBASE and the Cochrane Central Register of Controlled Trials was conducted on 4 April 2023. Included studies used ML models for clinical prediction in hip arthroscopy. Data on study design, outcomes, model development, validation and adherence to TRIPOD guidelines were extracted. Results Seventeen studies involving 37,549 patients (52.8% female; mean age 34.4 ± 6.1 years) were included. ML was used to predict clinical outcomes (47.1%), adverse events (47.1%) and resource utilisation (5.9%). The median area under the curve was 0.66 (range 0.5-0.94) for clinical outcomes and 0.7 (range 0.6-0.9) for adverse events. Resource utilisation was reported with a root-mean-square error of $3800, a logarithmic error of 0.2, and R^2 of 0.8. Only four studies compared ML to traditional regression; two found no difference, while two favoured ML. On average, studies met 12.3 of the 22 TRIPOD criteria, with only 29.4% meeting more than two-thirds. Conclusion ML models in hip arthroscopy show variable performance in predicting outcomes and complications.

Furthermore, most studies comparing ML models to traditional regression did not report any significant difference. Lastly, variability in adherence to TRIPOD guidelines highlights the need for future studies to improve transparency and reporting in the development of ML algorithms for hip arthroscopy. Given the potential of ML models in enhancing clinical decision-making for hip arthroscopy, along with the findings of the current study, their application should be used with caution. Level of evidence Level IV, systematic review of level II-IV studies.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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